UNIVERSITY OF PERADENIYA
DEPARTMENT OF PHYSICS

END SEMESTER EXAMINATION
SEMESTER II - 2018/2019

PH 102 - General Physics II

Time allowed: ***Three hours***

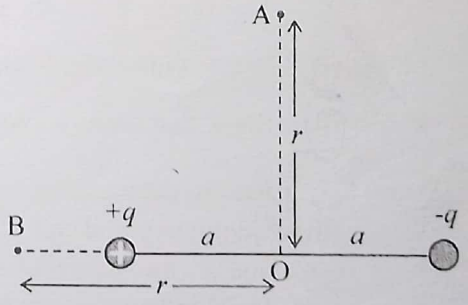
Answer **FIVE** questions selecting **TWO** from section-**A**, **TWO** from section-**B** **and ONE** from section-**C**.
(Calculators will be provided)

Electron charge (e)	$= 1.60 \times 10^{-19}$ C
Mass of electron (m_e)	$= 9.11 \times 10^{-31}$ kg
Mass of proton (m_p)	$= 1.67 \times 10^{-27}$ kg
Permittivity of free space (ε_o)	$= 8.85 \times 10^{-12}$ F m^{-1}
Permeability of free space (μ_o)	$= 4\pi \times 10^{-7}$ H m^{-1}
Planck's constant (h)	$= 6.63 \times 10^{-34}$ J s
Speed of light (c)	$= 3.00 \times 10^{8}$ m s^{-1}
Stefan-Boltzmann constant (σ)	$= 5.67 \times 10^{-8}$ W m^{-2} K^{-4}
Avogadro number (N_A)	$= 6.02 \times 10^{23}$ mol^{-1}

Section-A

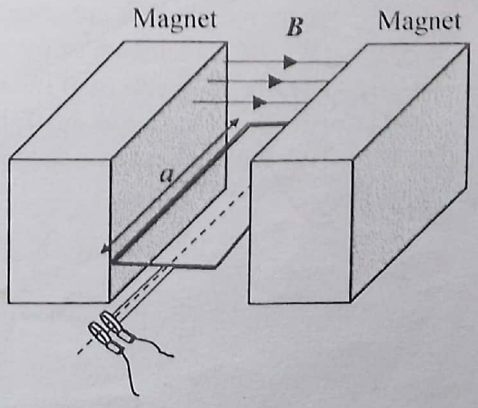
1. (a) (i) Clearly identifying the symbols used, write down an expression for the electric field intensity (E) at distance r from a point charge Q.
(ii) Define the electric potential and obtain an expression for potential (V) at distance r from a point charge Q.

(b) A dipole shown in the figure consists $+q$ and $-q$ charges separated by distance $2a$. O is the midpoint of the dipole. Point A is on the perpendicular bisector at distance r from O. Point B is on the axis at distance r from O.
(i) Obtain expressions for E_A and V_A at point A.
(ii) Also obtain expressions for E_B and V_B at point B.
(iii) Determine the value of r if the magnitude if magnitudes of $E_A = E_B$.



2. (a) Write down the laws of electromagnetic induction.

(b) A rectangular coil of length a and width b has N number of turns. It is pivoted about an axis through the center and parallel to the length as shown in the figure.



[Continued.....]

[.....continued]

(i) A uniform magnetic field of flux density $\boldsymbol{B}$ is applied perpendicular to the axis of rotation. If the plane of the coil makes an angle $(90°-\theta)$ with the magnetic field and a steady current I passes through the coil, obtain an expression for the torque on the coil.

(ii) Instead of passing a current, if the coil is rotated at constant angular velocity ω about the axis, obtain an expression for the electromotive force (e.m.f.) produces in the coil.

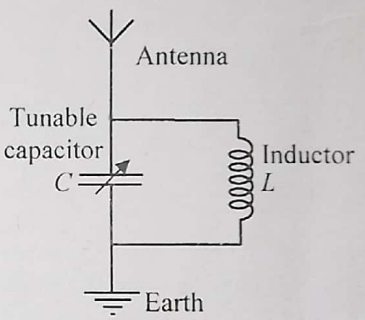
(iii) Plot the variation of above e.m.f. with time.

3. (a) A charged capacitor of capacitance C, an inductor of inductance L and a key are connected in series. Neglect the resistance of the circuit.

(i) If the switch is closed at time $t = 0$, obtain expressions for charge $q(t)$ in the capacitor and current $I(t)$ in the circuit at time t.

(ii) Sketh the variations of $q(t)$ and $I(t)$ as function of time (t) in a same graph.

(b) An antenna circuit, tuned to receive a radio signal, consists of a capacitor of capacitance C and an inductor of inductance L, as shown in the figure. Neglect the resistance of the circuit.



(i) If the inductance $L = 5.0$ μH, find the value of capacitance C to receive a radio signal of frequency 560 kHz.

(ii) If the total energy of the C-L circuit is 0.2 μJ, find the maximum current in the circuit and the maximum voltage across the capacitor.

SECTION B

4. (a) (i) What is meant by Compton effect?

(ii) Show that the Compton shift is given by $\lambda' - \lambda = \frac{h}{m_e c}(1 - \cos\theta)$. Identify the symbols used in the equation.

(iii) Qualitatively sketch the graph of intensity versus wavelength indicating peaks of λ and λ' for scattering angles $\theta = 0°$, 45°, 90° and 180°.

(iv) How do you explain the presence of the original wavelength λ for $\theta = 180°$?

(b) X-ray photons of wavelength 1.54×10^{-11} m are incident on a target and the Compton scattered photons are observed at 90°. Calculate

(i) the wavelength of the scattered photons.

(ii) the momentum of the incident photons and the scattered photons.

(iii) the kinetic energy of the scattered electrons and

(iv) the momentum (magnitude and direction) of the scattered electrons.

5. (a) (i) Clearly stating the assumptions deduce Einstein's photoelectric equation.

(ii) In a photoelectric experiment sodium surface is illuminated by a beam of electromagnetic radiation. If the work function of sodium is 2.3 eV, what is the largest wavelength that can emit photoelectrons?

(iii) Find the stopping potential when radiation of wavelength 450 nm is used in the above experiment.

(b) (i) State de Broglie Hypothesis.

(ii) What is the de Broglie wavelength associated with an electron having kinetic energy of 100 keV?

(iii) Explain how you would prove the wave nature of a beam of electrons by a simple experiment.

(iv) Calculate the wavelength associated with a proton with same kinetic energy.

(v) Can you use the same experiment described for part c to demonstrate the wave nature of the proton beam? Give reasons.

(c) (i) Write down an expression for electron energy of the nth level of a hydrogen atom based on the Bohr model.

(ii) Calculate the shortest and the longest wavelength associated with Balmer series of H atom.

(iii) Which range of the electromagnetic spectra that these two wavelength belong?

6. (a) (i) Write down the law of radioactive decay.

(ii) The number of radioactive nuclei in a sample at time $t = 0$ is N_o. Obtain an expression for the number of radioactive nuclei N after time t. Identify the symbols used in your expression.

(iii) Define half-life $T_{1/2}$ and obtain an expression for $T_{1/2}$ in term of the constant used in the above expression.

(iv) Radioactive $^{137}_{55}Cs$ has a half-life of 30.2 years. How many years will it take to reduce activity of Cs-137 to 10% of its initial value?

(b) Consider the following nuclear fission reaction in a nuclear reactor.

$$^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{137}_{55}Cs + {}^{96}_{a}X + b\,{}^{1}_{0}n$$

(i) Find the values of a and b.

(ii) If $^{96}X = 95.934$ u , p = 1.00728 u, n = 1.00867 u and 1 u = 931.5 MeV/c^2, find the binding energy per nucleon ($\bar{B}$) of the ^{96}X nucleus.

(iii) Calculate the energy released (in joule) from 50 mg of pure U-235.

Mass of the ^{235}U nucleus = 235.044 u

Mass of the ^{137}Cs nucleus = 136.907 u

(c) (i) Distinguish fermions and bosons?

(ii) Give the two types of elementary fermions and their flavors (types).

(iii) What are most stable baryons in the universe? Write down their constituents.

1 g = 6.022×10^{23} u

1 mg = 6.022×10^{20} u.

SECTION C

7. (a) (i) Write down the intermolecular bonding type present in the following compounds and elemental solids.

Water	Liquid methane	Tungsten	Diamond
MgO	Silicon	Liquid HF	Brass

(ii) Name the three types of dispersion forces used to describe the Van der Waals interactions among atoms/molecules and briefly explain them.

(b) The resultant potential function of a diatomic molecule is given by $V(r) = \frac{A}{r^9} - \frac{B}{r}$, where r is the distance between the atoms and A, B are constants.

(i) Identify the attractive and repulsive potentials.

(ii) Qualitatively sketch the above two potentials vs r using dotted lines and sketch the resultant $V(r)$ using a solid line in the same diagram. Name the three curves.

(iii) Calculate the values of A and B (in nm and eV) for a diatomic molecule of binding energy of 4.26 eV and the bond length of 0.236 nm.

(iv) Calculate the effective force constant of the bond between the two atoms of part (iii).

8. (a) (i) The rotational energy of a diatomic molecule is given by $E_j = \frac{\hbar^2}{2I} j(j+1)$. Identify j and I.

(ii) Show that the quantity $\hbar^2/I$ known as "characteristic rotational energy" has the unit of energy.

(iii) The characteristic rotational energy of N_2 is 2.48×10^{-4} eV and the atomic mass of N is 14.01 g mol^{-1}. Calculate the bond length of N_2.

(iv) The emitted wavelengths of photons for the transitions between the rotational energy levels $j = 1$ and $j = 0$ of two gas samples (CO and NO) are 4.5 mm and 2.6 mm. If the bond lengths of the two molecules are the same, identify their respective emission wavelengths. Qualitatively explain your answer. (Mass numbers of C = 12 and N =14)

(b) (i) Vibrations associated with H_2 molecules emit infrared radiation of 2300 nm. Calculate the energy difference between two vibrational levels.

(ii) Calculate the ground state energy of the vibrational levels of H_2 molecule.

(iii) Comment about the stretching vibrational frequencies of C=C and C-C bonds.

(iv) Give an application of atomic vibrations.

**********END**********